

Path-integral formulation of quantum mechanics

In the usual formulation of quantum mechanics, the quantities q and p are replaced by operators which obey Heisenberg commutation relations:

The mathematics one invokes is that of operators in Hilbert space.

- The path-integral formulation of quantum mechanics, on the other hand, is based directly on the notion of a propagator $K(q_f t_f; q_i t_i)$.
- Given a wave function $\Psi(q_i, t_i)$ at time t_i , the propagator gives the corresponding wave function at a later time t_f by an appeal to Huygens' principle:

$$\Psi(q_f, t_f) = \int K(q_f t_f; q_i t_i) \Psi(q_i, t_i) dq_i. \quad (291)$$

→ For simplicity we consider only one spatial dimension.

- The equation (291) is quite general and merely

expresses causality.

- According to the usual interpretation of quantum mechanics, $\psi(q_f, t_f)$ is the probability amplitude that the particle is at the point q_f at the time t_f , so $K(q_f, t_f; q_i, t_i)$ is the probability amplitude for a transition from q_i at time t_i to q_f at time t_f .

The probability that it is observed at q_f at time t_f is

$$P(q_f, t_f; q_i, t_i) = |K(q_f, t_f; q_i, t_i)|^2 \quad (292)$$

- Let us divide the time interval between t_i and t_f into two, with t as the intermediate time, and q the intermediate point in space, as shown in Fig. 1.

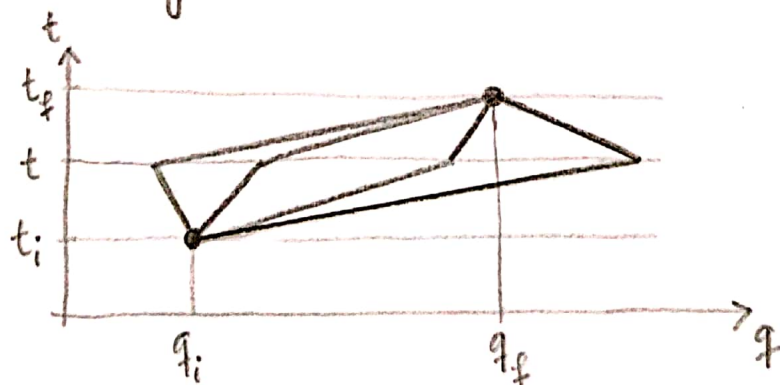


FIG. 1

[Propagation of a particle from (q_i, t_i) to (q_f, t_f)
via an intermediate position (q, t)]

Repeated applications of (291) gives

$$\psi(q_f, t_f) = \iint K(q_f, t_f; q, t) K(q, t; q_i, t_i) \psi(q_i, t_i) dq_i dq \quad (293)$$

From (291) we have $\psi(q_f, t_f) = \int K(q_f, t_f; q, t) \psi(q, t) dq$
and $\psi(q, t) = \int K(q, t; q_i, t_i) \psi(q_i, t_i) dq_i$

From (292) and (293):

$$K(q_f, t_f; q_i, t_i) = \int K(q_f, t_f; q, t) K(q, t; q_i, t_i) dq, \quad (294)$$

so the transition from (q_i, t_i) to (q_f, t_f) may be regarded as the result of transition from (q_i, t_i) to all available intermediate points q followed by transitions from (q, t) to (q_f, t_f) .

• As a simple and familiar illustration of this, consider the 2-slit experiment with electrons, show in Fig. 2

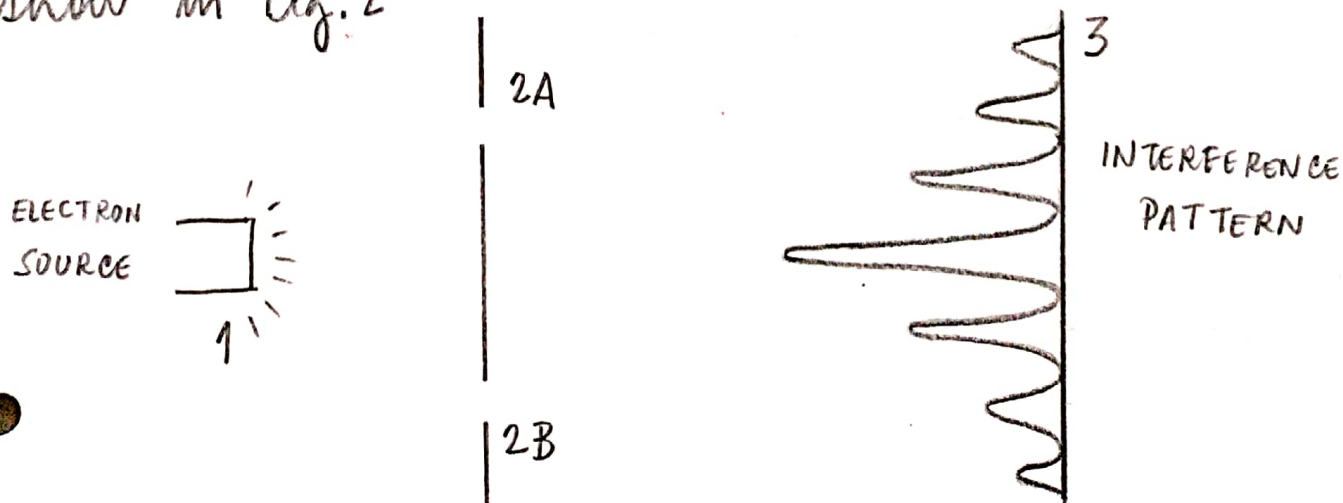


FIG. 2: THE 2-SLIT EXPERIMENT

Denote by $K(2A; 1)$ the probability amplitude that the electron passes from the source 1 to the hole 2A, and by $K(3; 2A)$ the amplitude that it passes from the hole 2A to the detectors 3, and so on.

Equation (294) then gives

$$K(3; 1) = K(3; 2A)K(2A; 1) + K(3; 2B)K(2B; 1) \quad (295)$$

and the intensity pattern of the screen 3 is given by the probability

$$P(3; 1) = |K(3; 1)|^2 \quad (296)$$

which will clearly contain interference terms, characteristic of the quantum theory.

- Note that we cannot say "the electron travelled either through hole A or hole B" - it travelled, in a sense, over both paths (if not detected at the slits).

- this notion of all possible paths is important in the path integral formalism

- for an excellent discussion of the 2-slit experiment, see Feynman, Leighton & Sands, "Lectures on Physics", volume III.

In this slightly different notation we have

$$K(q_f t_f; q_i t_i) = \langle q_f t_f | q_i t_i \rangle \quad (297)$$

→ The propagator K summarises the quantum mechanics of the system.

In the usual formulation of quantum mechanics, given an initial wave function, one can find the final wave function by solving the time-dependent Schrödinger equation.

→ In this formulation, however, the propagator gives the solution directly. The idea now is to express $\langle q_f t_f | q_i t_i \rangle$ as a path integral.

• Let us split the time interval between t_i and t_f into $(n+1)$ equal pieces τ , as in Figure 3.

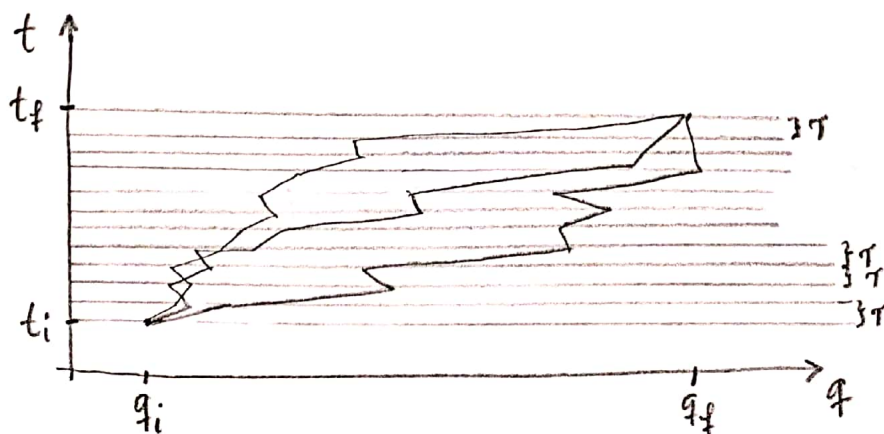


FIG. 3: Propagation over many paths from (q_i, t_i) to (q_f, t_f)

Equation (294) now becomes

$$\langle q_f t_f | q_i t_i \rangle = \int \dots \int dq_1 dq_2 \dots dq_m \langle q_f t_f | q_m t_m \rangle \langle q_m t_m | q_{m-1} t_{m-1} \rangle \dots \times \langle q_1 t_1 | q_i t_i \rangle. \quad (298)$$

• In fact, for $m=1$:

$$\langle q_f t_f | q_i t_i \rangle = \int \langle q_f t_f | q_1 t_1 \rangle \langle q_1 t_1 | q_i t_i \rangle dq_1$$

where

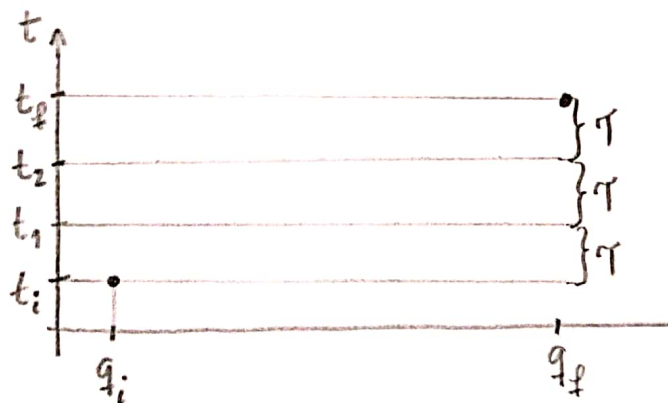
$$K(q_f t_f; q_i t_i) = \int K(q_f t_f; q_1 t_1) K(q_1 t_1; q_i t_i).$$

→ it is the previous result (293) and (294).

• For $m=2$:

$$\langle q_f t_f | q_i t_i \rangle = \iiint \langle q_f t_f | q_2 t_2 \rangle \langle q_2 t_2 | q_1 t_1 \rangle \langle q_1 t_1 | q_i t_i \rangle dq_1 dq_2,$$

where



- The integral in (296) is taken over all possible "trajectories"; they are not trajectories in the normal sense, since each segment $(q_j, t_j; q_{j-1}, t_{j-1})$ may be subdivided into smaller segments, so there is no derivative.
 $\rightarrow$ the paths are really Markov chains.

- Let us calculate the propagator over a small segment in the path integral:

$$\langle q_{j+1}, t_{j+1} | q_j, t_j \rangle = \langle q_{j+1} | e^{-iHt_{j+1}/\hbar} e^{iHt_j/\hbar} | q_j \rangle$$

$$= \langle q_{j+1} | e^{-iH\tau/\hbar} | q_j \rangle$$

$$= \langle q_{j+1} | 1 - \frac{i}{\hbar} H\tau + \mathcal{O}(\tau^2) | q_j \rangle$$

$$= \langle q_{j+1} | 1 | q_j \rangle - \frac{i\tau}{\hbar} \langle q_{j+1} | H | q_j \rangle + \cancel{\langle q_{j+1} | \mathcal{O}(\tau^2) | q_j \rangle}$$

$$= \delta(q_{j+1} - q_j) - \frac{i\tau}{\hbar} \langle q_{j+1} | H | q_j \rangle,$$

where $\tau = t_{j+1} - t_j$.

Remembering that $\delta(x-x') = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \cdot e^{ik(x-x')}$,

we have:

$$\begin{aligned}
 \langle q_{j+1} t_{j+1} | q_j t_j \rangle &= \delta(q_{j+1} - q_j) - \frac{i\tau}{\hbar} \langle q_{j+1} | H | q_j \rangle \\
 &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp \exp\left[\frac{i}{\hbar} p (q_{j+1} - q_j)\right] - \frac{i\tau}{\hbar} \langle q_{j+1} | H | q_j \rangle. \quad (299)
 \end{aligned}$$

- The Hamiltonian H is a function of the operators p and q .

In the special case where H is of the form

$$H = \frac{p^2}{2m} + V(q), \quad (300)$$

the matrix element in (299) may be calculated.
We have:

$$\begin{aligned}
 \left\langle q_{j+1} \left| \frac{p^2}{2m} \right| q_j \right\rangle &= \left\langle q_{j+1} \left| \mathbb{1} \frac{p^2}{2m} \mathbb{1} \right| q_j \right\rangle \\
 &= \left\langle q_{j+1} \left| \int dp' |p'\rangle \langle p'| \frac{p^2}{2m} \int dp |p\rangle \langle p| \right| q_j \right\rangle \\
 &= \iint dp' dp \langle q_{j+1} | p' \rangle \left\langle p' \left| \frac{p^2}{2m} \right| p \right\rangle \langle p | q_j \rangle. \quad (301)
 \end{aligned}$$

Now substitute

$$\langle q_{j+1} | p' \rangle = \sqrt{\frac{1}{2\pi\hbar}} \exp\left[\frac{i p' q_{j+1}}{\hbar}\right], \text{ giving}$$

$$\begin{aligned}
 \left\langle q_{j+1} \left| \frac{p^2}{2m} \right| q_j \right\rangle &= \iint \frac{dp' dp}{\sqrt{2\pi\hbar}} e^{i p' q_{j+1}/\hbar} \frac{p^2}{2m} \langle p' | p \rangle \frac{1}{\sqrt{2\pi\hbar}} e^{-i p q_j/\hbar} \\
 &= \iint \frac{dp' dp}{2\pi\hbar} \exp \left[\frac{i}{\hbar} (p' q_{j+1} - p q_j) \right] \frac{p^2}{2m} \delta(p - p') \\
 &= \int \frac{dp}{\hbar} \exp \left[\frac{i}{\hbar} p (q_{j+1} - q_j) \right] \frac{p^2}{2m}. \quad (302)
 \end{aligned}$$

• It is to be noted here that p^2 on the left-hand side of (302) is an operator, whereas on the right-hand side it is a number.

• In a similar way,

$$\begin{aligned}
 \langle q_{j+1} | V(q) | q_j \rangle &= V \left(\frac{q_{j+1} + q_j}{2} \right) \langle q_{j+1} | q_j \rangle \\
 &= V \left(\frac{q_{j+1} + q_j}{2} \right) \delta(q_{j+1} - q_j) \\
 &= \int \frac{dp}{\hbar} \exp \left[\frac{i}{\hbar} p (q_{j+1} - q_j) \right] V(\bar{q}_j), \quad (303)
 \end{aligned}$$

where $\bar{q}_j \equiv \frac{1}{2} (q_j + q_{j+1})$.

→ in order to obtain (303) I used the "Weyl ordering".

• See, for example, "An introduction to the functional formulation of Quantum Mechanics", Horacio O. Girotti.

Putting (302) and (303) together,

$$\langle q_{j+1} | H | q_j \rangle = \int \frac{dp}{h} \exp \left[\frac{i}{h} p (q_{j+1} - q_j) \right] H(p, \bar{q}_j) \quad (304)$$

and, from (299),

$$\begin{aligned} \langle q_{j+1} t_{j+1} | q_j t_j \rangle &= \frac{1}{h} \int dp_j \exp \left[\frac{i}{h} p (q_{j+1} - q_j) \right] - \frac{i\tau}{h} \langle q_{j+1} | H | q_j \rangle \\ &= \frac{1}{h} \int dp \exp \left[\frac{i}{h} p (q_{j+1} - q_j) \right] - \frac{i\tau}{h} \int \frac{dp}{h} \exp \left[\frac{i}{h} p (q_{j+1} - q_j) \right] H(p, \bar{q}_j) \\ &= \frac{1}{h} \int dp \exp \left[\frac{i}{h} p (q_{j+1} - q_j) \right] \left\{ 1 - \frac{i\tau}{h} H(p, \bar{q}_j) \right\} \\ &= \frac{1}{h} \int dp_j \exp \left[\frac{i}{h} p (q_{j+1} - q_j) \right] \left\{ e^{-iH\tau/h} \right\} \\ &= \frac{1}{h} \int dp_j \exp \left[\frac{i}{h} p (q_{j+1} - q_j) - iH\frac{\tau}{h} \right] \\ &= \frac{1}{h} \int dp_j \exp \left\{ \frac{i}{h} \left[p_j (q_{j+1} - q_j) - \tau H(p_j, \bar{q}_j) \right] \right\}, \end{aligned} \quad (305)$$

where p_j is the momentum between t_j and t_{j+1} , or, equivalently, q_j and q_{j+1} .

→ This gives the propagator over a segment of one possible path.

- The full propagator is got by substituting this into (298), giving, in the continuum limit (where p_j is the momentum along the path between q_j and q_{j+1}),

$$\langle q_f t_f | q_i t_i \rangle = \lim_{m \rightarrow \infty} \int \prod_{j=1}^m dq_j \prod_{j=0}^m \frac{dp_j}{h} \exp \left\{ \frac{i}{\hbar} \sum_{j=0}^m [p_j (q_{j+1} - q_j) - \tau H(p_j, \bar{q}_j)] \right\} \quad (306)$$

with $q_0 = q_i$, $q_{m+1} = q_f$.

- This may be written in the symbolic form

$$\langle q_f t_f | q_i t_i \rangle = \int \frac{\mathcal{D}q \mathcal{D}p}{h} \exp \frac{i}{\hbar} \left[\int_{t_i}^{t_f} dt [p \dot{q} - H(p, q)] \right] \quad (307)$$

with $q(t_i) = q_i$, $q(t_f) = q_f$.

- In the continuum limit q becomes a function of t , and the integral is a "functional integral", an integral over all functions.
→ it is infinite-dimensional.

- Expression (307) is the path integral expression for the transition amplitude from (q_i, t_i) to (q_f, t_f) .
→ each function $q(t)$ and $p(t)$ defines a path in phase space.

- In the path-integral formulation we have an explicit expression for the transition amplitude, which is clearly very well suited to scattering problems.

The quantities p and q occurring in the integral are classical quantities, not operators (c-numbers, not q-numbers).

It is not obvious, however, that infinite-dimensional integrals of this type are well-defined mathematically, that is, whether they converge; in other words, exist!

→ we shall assume that they do.

- There is another form for the transition amplitude, which holds when H is of the form $H = P^2/2m + V(q)$, since in that case we can perform the p -integration.

Equation (306) becomes

$$\langle q_f t_f | q_i t_i \rangle = \lim_{m \rightarrow \infty} \int \prod_{j=1}^m dq_j \prod_{j=0}^m \frac{dp_j}{h} \exp \left\{ \frac{i}{h} \sum_{j=0}^m \left[p_j (q_{j+1} - q_j) - \frac{p_j^2}{2m} \tau - V(q_j) \tau \right] \right\}. \quad (308)$$

As far as the p_j integration is concerned, this is of the same form as equation

$$\int_{-\infty}^{\infty} \exp(-ax^2 + bx + c) dx = \exp\left(\frac{b^2}{4a} + c\right) \sqrt{\frac{\pi}{a}}. \quad (309)$$

So we get

$$\langle q_f t_f | q_i t_i \rangle = \lim_{m \rightarrow \infty} \left(\frac{m}{i\hbar\tau} \right)^{\frac{(m+1)}{2}} \int \prod_{j=1}^m dq_j \exp \left\{ \frac{i\tau}{\hbar} \sum_{j=0}^m \left[\frac{m}{2} \left(\frac{q_{j+1} - q_j}{\tau} \right)^2 - V \right] \right\} \quad (310)$$

and hence, in the continuum limit

$$\langle q_f t_f | q_i t_i \rangle = N \int \mathcal{D}q \exp \left[\frac{i}{\hbar} \int_{t_1}^{t_2} L(q, \dot{q}) dt \right] \quad (311)$$

• where $L = T - V$, the classical Lagrangian.

In the $m \rightarrow \infty$, N becomes infinite, but this does not matter, since we shall always deal with normalized transition amplitudes.

- The integrand in (311) is the classical action $S = \int L dt$. We have proved this equation from the postulates of quantum mechanics, and by assuming that the Hamiltonian is of the form (300).

→ Feynman's original approach was to adopt (311) as a hypothesis and then prove the Schrödinger equation from it. The disadvantage of this approach is that (311) does not hold in general, since (300) does not.